Unit 2 Quiz A: Coaching Guide

Don't just check the letter. For each question, answer the 'Ask yourself' prompts BEFORE you read the steps.

Q	1	2	3	4	5	6	7	8	9	10	11	12	13
Key	**B**	**B**	**B**	**A**	**C**	**C**	**C**	**C**	**B**	**B**	**C**	**B**	**A**

- **Score it:** right + sure = solid. Right + guess = review. Wrong = full walkthrough + Go deeper.
- Re-attempt every miss on paper tomorrow, without looking. If you get it right then, it sticks.

Question 1

`2.1 Center of mass`

Three small objects sit on a light rod: 2.0 kg at x = 0, 3.0 kg at x = 2.0 m, and 5.0 kg at x = 4.0 m. Where is the center of mass of the system?

ASK YOURSELF FIRST

- Before any math: is the center of mass left or right of the middle of the rod (2.0 m)? Which end is heavier?
- Every object gets a 'vote' on the location. What decides how loud its vote is?

STEP BY STEP

1. Write the definition: $x_{cm} = \Sigma(m \cdot x) / \Sigma m$.
2. Top: (2.0)(0) + (3.0)(2.0) + (5.0)(4.0) = 0 + 6 + 20 = 26 kg·m.
3. Bottom: total mass = 2.0 + 3.0 + 5.0 = 10 kg. Every mass goes in the bottom, including the one at x = 0.
4. x_{cm} = 26 / 10 = 2.6 m. Check: right of the middle, toward the 5 kg end, as predicted.

ANSWER B. x = 2.6 m

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** 2.0 m is the geometric middle. It ignores mass, as if all three objects weighed the same.
- **C:** 3.25 m = 26/8. The 2.0 kg object sits at x = 0, so it adds nothing to the top, but it still counts in the total mass.
- **D:** The center of mass is pulled toward the heaviest object, but it isn't AT the heaviest object.

GO DEEPER Where would you place an extra 10 kg mass to move the center of mass to exactly x = 2.0 m? (Set up (10x + 26)/20 = 2.0. Answer: x = 1.4 m.)

Question 2

`2.2 / 2.4 FBD + equilibrium`

A crate is dragged across a rough floor at CONSTANT velocity by a rope that pulls with tension T at angle θ above the horizontal. Which statement is correct?

ASK YOURSELF FIRST

- 'Constant velocity' is a gift. What does it tell you about the net force in the x-direction AND in the y-direction?
- The rope is tilted. Does it only pull forward, or does part of it pull up too? What does that do to how hard the floor has to push?

STEP BY STEP

1. Draw the FBD: F_g down, F_N up, friction backward, T tilted up and forward.
2. Split T: forward part T cos θ, upward part T sin θ.
3. x: constant v → $\Sigma F_x = 0$ → T cos θ − f = 0 → f = T cos θ.
4. y: $\Sigma F_y = 0$ → F_N + T sin θ − mg = 0 → F_N = mg − T sin θ, which is less than mg. The rope carries part of the weight.

ANSWER B. F_N < mg and f = T cos θ

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Forgets that the rope is tilted. Only the horizontal part of T balances friction, and the vertical part lightens the load on the floor.
- **C:** Sign slip: the rope pulls UP, so the floor pushes LESS.
- **D:** The 'you need extra force to keep it moving' myth. Constant velocity means the forces balance exactly.

GO DEEPER Pushing a lawnmower: the handle pushes DOWN at angle θ below horizontal. Write F_N for that case. Why is pushing at an angle harder work than pulling at the same angle? (F_N = mg + F sin θ, so friction is bigger.)

Question 3

2.3 Newton's 3rd law

Alex (50 kg) and Blake (100 kg) sit on rolling office chairs facing each other. Alex pushes on Blake's shoulders and they roll apart. While the hands are in contact, which is true?

ASK YOURSELF FIRST

- Blake isn't pushing at all, so can there still be a force from Blake on Alex?
- Two separate questions: (1) how do the forces compare, and (2) how do the accelerations compare? Which law answers each?

STEP BY STEP

1. 3rd law: the force of Alex on Blake and the force of Blake on Alex are a pair, so they're equal in size and opposite in direction. Who 'started it' doesn't matter.
2. 2nd law, applied to each person: a = F / m. Same F.
3. a_{Alex} = F / 50 and a_{Blake} = F / 100, so Alex's acceleration is 2 times Blake's.

ANSWER B. Equal forces, and Alex accelerates twice as much

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** 'The one doing the pushing pushes harder' is the #1 3rd-law myth.
- **C:** Equal forces don't mean equal effects. Mass decides the acceleration.
- **D:** Mass changes the acceleration, not the size of the interaction force.

GO DEEPER If Blake pushes instead, does anything in your answer change? (No. The forces are still equal, and Alex's acceleration is still twice Blake's.)

Question 4

2.5 Newton's 2nd law + graphs

The graph shows the velocity of an elevator (up is positive). A rider stands on a scale. Rank the scale reading during intervals I, II and III.

ASK YOURSELF FIRST

- The scale doesn't measure your weight. It measures the normal force. What controls the normal force: velocity or acceleration?
- How do you read acceleration off a v–t graph?

STEP BY STEP

1. Slope of the v–t graph = acceleration. I: +2 m/s² (up). II: 0. III: –2 m/s² (down).
2. Rider's FBD: F_N up, mg down. 2nd law with up as +: $F_N - mg = ma \rightarrow F_N = m(g + a)$.
3. I: m(g + 2). II: mg. III: m(g – 2). So I > II > III.

ANSWER A. I > II > III

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** Comes from thinking 'it's always going up, so nothing changes'. The scale reacts to acceleration, not to direction of travel.
- **D:** Picks the interval with the largest velocity. Velocity doesn't show up in $F_N = m(g + a)$ at all.

GO DEEPER For a 50 kg rider, compute all three readings. (590 N, 490 N, 390 N.) Then sketch the scale reading vs. time. It's a step graph, not a copy of the v–t graph.

Question 5

2.6 Gravitational field

Planet X has twice Earth's mass and twice Earth's radius. What is the gravitational field strength at its surface? (Earth: g = 9.8 N/kg)

ASK YOURSELF FIRST

- Write g in symbols first. Which variable helps (makes g bigger), and which hurts?
- Radius is squared. So which change wins, the mass doubling or the radius doubling?

STEP BY STEP

1. $g = GM / R^2$.
2. New $g = G(2M) / (2R)^2 = (2/4) \cdot GM/R^2 = \frac{1}{2}\, g_{Earth}$.
3. ½ × 9.8 = 4.9 N/kg.

ANSWER C. 4.9 N/kg

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Doubles the mass but forgets the radius.
- **B:** 'Twice the mass, twice the radius, so they cancel.' The radius is squared, so it doesn't cancel.
- **D:** Squared the mass too. Only the distance is squared.

GO DEEPER What radius would Planet X need for g to be exactly 9.8 N/kg again? ($2M/R^2 = M/R_E^2$, so $R = \sqrt{2}\, R_E \approx 1.41\, R_E$.) This 'factor of change' style is very common on the AP exam.

Question 6

2.6 Apparent weight

Astronauts float inside the International Space Station, about 400 km above Earth's surface (Earth's radius is about 6400 km). Which explanation is correct?

ASK YOURSELF FIRST

- Estimate: 400 km is how big compared with 6400 km? Does that make gravity much weaker, or only a little?
- What does a bathroom scale actually measure? What would have to push on the astronaut for them to 'feel' weight?

STEP BY STEP

1. Field at the ISS: $g \times (6400 / 6800)^2 \approx 0.89\, g \approx 8.7$ N/kg. Gravity is still almost full strength.
2. Gravity is the only force on both the station and the astronaut, so both accelerate toward Earth at the same rate, 8.7 m/s².
3. Since they accelerate together, the floor never needs to push the astronaut. $F_N = 0$, and 'weightless' really means no normal force (apparent weight = 0).
4. They don't hit Earth because they move sideways fast enough that they keep falling 'around' it. That's an orbit.

ANSWER C. Both in free fall together, so the normal force is zero

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** If there were no gravity, the station would fly off in a straight line (1st law). Gravity is what bends its path into a circle.
- **B:** The drop is only about 11%, not 90%. Squaring 6400/6800 doesn't shrink it much.
- **D:** Centripetal isn't an extra force. Here gravity IS the centripetal force, so nothing gets cancelled.

GO DEEPER Connect it: an elevator whose cable snaps. What does the scale read while it falls? (Zero, for the same reason.)

Question 7

2.6 Universal gravitation

A satellite weighs W on Earth's surface. It is launched to an altitude ABOVE THE SURFACE equal to one Earth radius. What is the gravitational force on it there?

ASK YOURSELF FIRST

- In $F = Gm_1m_2/r^2$, is r measured from the surface or from the center?
- At the surface, how far is the satellite from Earth's center? And at the new position?

STEP BY STEP

1. At the surface, $r = R$. At an altitude of one radius, $r = R + R = 2R$.
2. The distance from the center doubles, so the force is $1/2^2 = 1/4$ as big.

3. F = W / 4.

ANSWER C. W / 4

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Assumes the satellite's weight doesn't change, as if 'weight' were fixed like mass. Mass is fixed. Weight is not.
- **B:** Treats the force as 1/r instead of $1/r^2$.
- **D:** Gravity never reaches zero, at any distance.

GO DEEPER At what altitude would its weight be W/9? (r = 3R, so the altitude is 2R.)

Question 8

2.7 Static friction

A 5.0 kg box rests on a floor ($\mu_s = 0.50$, $\mu_k = 0.30$). You push it horizontally with 20 N and it does not move. What is the friction force on the box?

ASK YOURSELF FIRST

- The box doesn't move. What does that force the net force to be?
- Is $\mu_s F_N$ the friction force, or the LIMIT on the friction force?

STEP BY STEP

1. The box is at rest, so a = 0 and ΣF = 0. Friction must exactly balance your 20 N push.
2. Check that static friction CAN supply 20 N: the maximum is $\mu_s mg = 0.50 \times 5.0 \times 9.8 = 24.5$ N. 20 < 24.5, so yes.
3. Static friction = 20 N, pointing opposite your push.

ANSWER C. 20 N

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** No motion doesn't mean no friction. Friction is what's keeping it still.
- **B:** 14.7 N is kinetic friction, which only applies once the box is sliding.
- **D:** 24.5 N is the MAXIMUM static friction. Static friction is only as big as it needs to be, up to that limit. With 24.5 N against your 20 N push, the box would accelerate toward you!

GO DEEPER You increase the push to just over 24.5 N and the box starts sliding. If you keep pushing with 24.5 N, what is the acceleration? ((24.5 – 14.7)/5.0 = 1.96 m/s². That's why boxes 'jump' when they finally break free.)

Question 9

2.7 Friction on an incline

A block of mass m rests, not moving, on a rough ramp tilted at angle θ. The coefficient of static friction is μ_s. What is the magnitude of the friction force on the block?

ASK YOURSELF FIRST

- Tilt your axes: one along the ramp and one perpendicular to it. Which part of gravity tries to slide the block down?
- Is this block at the edge of slipping? Does the question say so?

STEP BY STEP

1. Along the ramp, the part of gravity pulling down the slope is mg sin θ.
2. The block is at rest, so ΣF along the ramp = 0: f – mg sin θ = 0.
3. f = mg sin θ. μ_s doesn't appear, because the block isn't at the edge of slipping.

ANSWER B. mg sin θ

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** $\mu_s$mg cos θ is the MAXIMUM static friction. You only use it when the block is 'just about to slip'.
- **C:** mg cos θ is the perpendicular part of gravity. It equals F_N, not friction.
- **D:** Mixes two formulas together. Check it with units and limits: at θ = 0 friction should be 0 ✓, but with μ_s = 0 it would also predict 0 friction for a block that is clearly at rest.

GO DEEPER Slowly tilt the ramp steeper. At what angle does the block start to slip? (mg sin θ = $\mu_s$mg cos θ, so tan θ = μ_s. Notice that the mass cancels!)

Question 10

2.5 Connected systems

A 3.0 kg block and a 5.0 kg block hang from opposite ends of a light string over an ideal pulley and are released. What is the tension in the string?

ASK YOURSELF FIRST

- Sanity check BEFORE the math: the 3 kg block accelerates upward. Must T be more or less than its weight (29.4 N)?
- The 5 kg block accelerates downward. Must T be more or less than ITS weight (49 N)? So where does T have to land?

STEP BY STEP

1. System (both blocks, string internal): the net force is (5.0 – 3.0)g = 19.6 N, pulling on 8.0 kg total, so a = 19.6 / 8.0 = 2.45 m/s².
2. Isolate the 3.0 kg block (it accelerates up): T – m_1g = m_1a → T = m_1(g + a) = 3.0(9.8 + 2.45).
3. T = 36.75 ≈ 36.8 N. Between 29.4 and 49 ✓.
4. Double-check with the 5 kg block: m_2g – T = m_2a → 49 – 36.75 = 12.25 = 5.0 × 2.45 ✓.

ANSWER B. 36.8 N

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** 29.4 N is the weight of the light block. That would mean a = 0.
- **C:** 39.2 N is the average of the two weights. It's close but not right. Averaging isn't physics, so derive it.
- **D:** 49 N is the heavy block's weight. Then the heavy block couldn't accelerate down.

GO DEEPER Derive T in symbols: $T = 2m_1m_2g / (m_1 + m_2)$. Check two limits: equal masses → $T = mg$ ✓. A huge m_2 → $T \to 2m_1g$. Why $2m_1g$? (The light block then accelerates up at almost g.)

Question 11

2.9 Circular motion

A car of mass m drives over the top of a rounded hill of radius r at speed v. What is the normal force from the road at the very top?

ASK YOURSELF FIRST

- At the top of the hill, which way does the car's acceleration point: up, down, or nowhere?
- Which is bigger then, the force pointing toward the center (down) or the one pointing away (up)?

STEP BY STEP

1. Moving along a circle, the acceleration points toward the center, which here is DOWN, with size v^2/r.
2. FBD: mg down, F_N up. Take toward the center (down) as +: $mg - F_N = mv^2/r$.
3. $F_N = mg - mv^2/r$. That's less than mg, which is why you feel light at the top of a hill.

ANSWER C. $mg - mv^2/r$

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Would mean no acceleration, but the car is changing direction.
- **B:** That's the bottom of a DIP, where the center is above you.
- **D:** Treats 'centripetal force' as its own force. It's the NET of mg and F_N.

GO DEEPER At what speed does the car just lose contact with the road ($F_N = 0$)? ($v = \sqrt{gr}$. For r = 20 m, about 14 m/s, or 31 mph.)

Question 12

2.9 Vertical circle · BOSS

A ball on a string moves in a VERTICAL circle at constant speed. Rank the string tension at the top (T_{top}), at the side where the string is horizontal (T_{side}), and at the bottom (T_{bot}).

ASK YOURSELF FIRST

- Constant speed means the net force toward the center has the same size everywhere. Which quantity is that?
- At each point, does gravity help the string (point toward the center), fight it (point away), or neither?

STEP BY STEP

1. Net force toward the center = mv^2/r, the same at all three points.
2. Top: tension AND gravity both point down, toward the center: $T + mg = mv^2/r \to T_{top} = mv^2/r - mg$. Gravity helps, so the string does less.
3. Side: gravity is sideways to the center line, so it doesn't count: $T_{side} = mv^2/r$.
4. Bottom: tension up (toward the center) and gravity down (away from it): $T - mg = mv^2/r \to T_{bot} = mv^2/r + mg$.
5. So $T_{bot} > T_{side} > T_{top}$. Strings break at the bottom.

ANSWER B. $T_{bot} > T_{side} > T_{top}$

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Upside-down thinking: at the top, gravity does part of the string's job.
- **C:** Constant speed makes the NET force constant in size, not the tension. Gravity's role changes around the circle.

GO DEEPER What is the slowest speed at the top that keeps the string tight? ($T_{top} = 0 \rightarrow v = \sqrt{(gr)}$.)

Question 13

2.8 Springs · optional: ask if springs are on the test

A 0.50 kg mass hangs at rest from a spring and stretches it 0.10 m. What is the spring constant, and how far would a 1.5 kg mass stretch the same spring?

ASK YOURSELF FIRST

- At rest, what two forces balance on the mass?
- Is stretch proportional to the load, or to the load squared?

STEP BY STEP

1. Equilibrium: $kx = mg \rightarrow k = (0.50)(9.8) / 0.10 = 49$ N/m.
2. Hooke's law is linear: 3 times the mass gives 3 times the stretch, so 0.30 m. (Check: $1.5 \times 9.8 / 49 = 0.30$ m ✓.)

ANSWER A. 49 N/m and 0.30 m

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** Used mass instead of weight (forgot g).
- **C:** Squared the factor of 3. Hooke's law is linear.

GO DEEPER Cut the spring in half. Is each half stiffer or softer? (Stiffer, k doubles: each half stretches half as much under the same load.)

Error log: Quiz A

For each miss or guess: what went wrong, using one of these types: **C** concept (didn't know the physics) · **R** representation (FBD/graph wrong) · **M** math/algebra · **F** FRQ wording (right idea, missing the words) · **T** time.

Q	Type	What I did vs. what I should have done	Re-try date / ✓

Score: _____ / 13 Sure-and-right: _____ Time used: _____